

Networking Fundamentals

A Project Report on Computer Network Basics and Practical Diagnostics

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Topic	Networking Fundamentals

1. Introduction

This report covers the fundamental concepts of computer networking, including the core devices, addressing systems, and terminology used to build and secure networks. It also documents practical verification of these concepts by examining the actual network configuration of the system used, along with connectivity tests performed using standard command-line networking tools.

2. Core Networking Concepts

2.1 What is a Computer Network?

A computer network is a group of two or more interconnected computers or devices that can communicate and share resources such as data, files, printers, and internet access with each other, typically using cables, radio waves, or other transmission media.

2.2 Client and Server

A **client** is a device or application that requests services or resources (such as web pages, files, or data) from another device on the network. A **server** is a device or program that provides those services or resources and responds to client requests. For example, a web browser (client) requests a page from a web server, which processes the request and sends back the page.

2.3 LAN and WAN

A **LAN (Local Area Network)** connects computers and devices within a small, limited geographic area, such as a single home, office, or building, and typically offers high data transfer speeds. A **WAN (Wide Area Network)**, on the other hand, spans a much larger geographic area, such as a country or the entire globe, and connects multiple LANs together. The internet is the largest example of a WAN, and WANs are generally slower than LANs since they often rely on leased telecommunication lines or satellite links.

2.4 Switches and Routers

A **switch** is a networking device used within a LAN to connect multiple devices, such as computers and printers, and forward data only to the specific device it is intended for, based on MAC addresses. This makes data transfer within the network efficient and reduces unnecessary traffic compared to a hub. A **router**, in contrast, connects different networks together, such as a LAN to the internet, and directs data packets between them based on IP addresses, determining the best path for data to travel from source to destination.

2.5 Firewalls

A firewall is a security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and untrusted external networks, such as the internet, blocking unauthorized access while allowing legitimate communication.

2.6 Addressing: IP and MAC

An **IP (Internet Protocol) address** is a unique numerical label assigned to each device connected to a network. It identifies the device and its location on the network, allowing data to be correctly routed to and from it; IP addresses can be IPv4 (e.g., 192.168.1.1) or IPv6. A **MAC (Media Access Control) address**, on the other hand, is a unique 48-bit hardware identifier assigned to a device's network interface card by the manufacturer. Unlike an IP address, a MAC address is generally fixed and is used for communication within a local network at the data link layer.

2.7 Default Gateway

A default gateway is a device, usually a router, that serves as the access point through which devices on a local network send data destined for other networks or the internet. If a device does not know how to reach a destination directly, it forwards the traffic to the default gateway.

2.8 DNS (Domain Name System)

DNS translates human-readable domain names, such as www.example.com, into machine-readable IP addresses. It acts like a phone book for the internet, allowing users to access websites using easy-to-remember names instead of numeric IP addresses.

2.9 Relevance of Networking to Cybersecurity

Networking knowledge is essential for cybersecurity because most attacks, threats, and vulnerabilities occur over networks. Understanding how data flows, how devices communicate, and how protocols work helps security professionals detect suspicious activity, configure firewalls and access controls, analyze traffic, identify intrusions, and secure systems effectively. Without networking fundamentals, it is difficult to understand how attacks like spoofing, sniffing, or man-in-the-middle attacks occur or how to defend against them.

3. My Network Configuration

The following details were obtained from the local system to verify the networking concepts discussed above in a practical setting.

Parameter	Value
Private IPv4 Address	10.105.195.218
Subnet Mask	255.255.255.0
Default Gateway	_____
DNS Server	_____

*Note: Default Gateway and DNS Server were not clearly visible in the captured ipconfig screenshot. These can be filled in after running **ipconfig /all** in Command Prompt, which lists the "Default Gateway" and "DNS Servers" fields under the active Wi-Fi adapter.*

4. Practical Verification

To confirm the network configuration and connectivity of the system, three standard command-line networking utilities were used: ipconfig, ping, and tracert. The outputs are documented below.

4.1 ipconfig

The ipconfig command was run to view the current IP configuration of the system.

```
Microsoft Windows [Version 10.0.17134.1]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::b775:b51b:c72a:89b3%15
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 8:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2402:8100:2ca0:8cda:b8cc:1b6f:414b:8085
    Temporary IPv6 Address. . . . . : 2402:8100:2ca0:8cda:b801:a866:e062:4cef
    Link-local IPv6 Address . . . . . : fe80::d7eb:a11:2cb8:850f%11
    IPv4 Address. . . . . : 10.105.195.218
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.56.1
```

Output shows IPv4 Address: 10.105.195.218, Subnet Mask: 255.255.255.0 on the active Wi-Fi adapter.

4.2 ping

The ping command was used to test connectivity to google.com and measure round-trip time.

```
C:\Users\user>ping google.com

Pinging google.com [2404:6800:4009:824::200e] with 32 bytes of data:
Reply from 2404:6800:4009:824::200e: time=81ms
Reply from 2404:6800:4009:824::200e: time=73ms
Reply from 2404:6800:4009:824::200e: time=68ms
Reply from 2404:6800:4009:824::200e: time=56ms

Ping statistics for 2404:6800:4009:824::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 56ms, Maximum = 81ms, Average = 69ms
```

4 packets sent, 4 received, 0% packet loss; average round-trip time approximately 69 ms.

4.3 tracert

The tracert command was used to trace the path taken by packets from this system to google.com.

```

C:\Users\user>tracert google.com

Tracing route to google.com [2404:6800:4000:1015::65]
over a maximum of 30 hops:

  1      3 ms      1 ms      1 ms  2402:8100:2ca0:8cda::5c
  2      *         *         *    Request timed out.
  3      *        54 ms      *    fd00:0:0:91::2d
  4     36 ms     36 ms     37 ms  fd00:0:16:16::52
  5      *         *         *    Request timed out.
  6    102 ms    227 ms     52 ms  2400:5200:a000::6:3f00:1ce
  7      *         *         *    Request timed out.
  8      *         *         *    Request timed out.
  9     77 ms     52 ms     56 ms  2402:8100:4000::60e
 10      *         *         *    Request timed out.

```

Output lists each hop along the route to google.com, along with response times for each hop.

5. Conclusion

This project covered the essential building blocks of computer networking, including network types, key devices, and addressing systems, along with a practical demonstration using ipconfig, ping, and tracert to inspect and verify real network settings and connectivity. This combination of theory and hands-on practice reinforces the understanding of how devices communicate on a network and lays the foundation for further study in network security.